

Module 5: Lesson 2

## LSTMs for predicting BTC prices



# Outline

- ▶ The prediction problem at hand
- ▶ EDA: BTC prices and returns
- ▶ LSTM network to predict BTC returns

# The prediction problem

As it is usual in finance, our interest one more time lies in predicting short-term future asset returns:

- ▶ Which asset? **Bitcoin (BTC)**
- ▶ Which returns? **25-hour ahead returns**
- ▶ Which inputs? Past cumulative returns for **1-hour, 10-hour, and 50-hour**
- ▶ Which period? **Jan-2021 to Sept-2022**

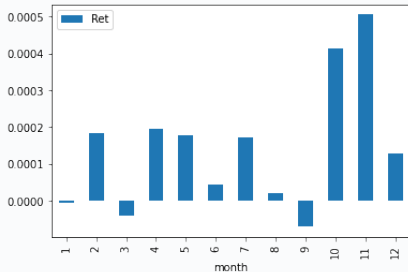
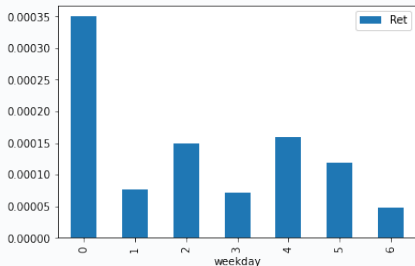
|              | date                | Ret       | Ret_10    | Ret_50   | Ret25     |
|--------------|---------------------|-----------|-----------|----------|-----------|
| <b>52634</b> | 2021-01-03 01:00:00 | 0.003812  | 0.000300  | 0.001236 | 0.000294  |
| <b>52635</b> | 2021-01-03 02:00:00 | 0.004669  | 0.000103  | 0.001244 | 0.000270  |
| <b>52636</b> | 2021-01-03 03:00:00 | -0.004577 | -0.000065 | 0.001066 | 0.000233  |
| <b>52637</b> | 2021-01-03 04:00:00 | 0.019479  | 0.000120  | 0.001334 | -0.000088 |
| <b>52638</b> | 2021-01-03 05:00:00 | 0.014644  | 0.000235  | 0.001445 | -0.000118 |

# Exploratory Data Analysis (EDA)

We will be working with what you may call high-frequency data on Bitcoin (BTC) prices. Since we are not so familiar with this data, we may find it useful to perform some simple Exploratory Data Analysis (EDA).

EDA is a particularly helpful way to learn more about the data throughout visualization. This way, we can make more informed choices about how to deal with outliers, the type of scaling needed, etc.

Here are some of the results from EDA:

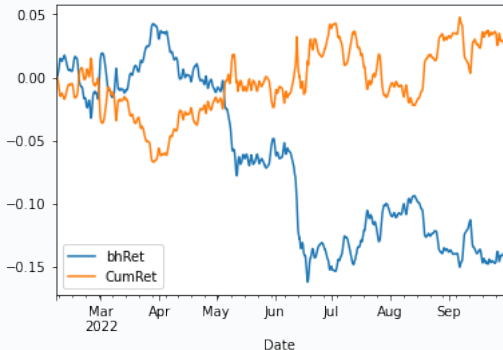


# LSTM model to predict BTC returns

In the notebook accompanying this Lesson, we will guide you through the development of an LSTM model. Here you have the summary of the model, as well as the out-of-sample performance of a backtest for the strategy using model predictions:

| Layer (type)        | Output Shape   | Param # |
|---------------------|----------------|---------|
| lstm (LSTM)         | (None, 30, 50) | 10800   |
| lstm_1 (LSTM)       | (None, 30, 50) | 20200   |
| dropout (Dropout)   | (None, 30, 50) | 0       |
| lstm_2 (LSTM)       | (None, 30, 50) | 20200   |
| dropout_1 (Dropout) | (None, 30, 50) | 0       |
| lstm_3 (LSTM)       | (None, 50)     | 20200   |
| dropout_2 (Dropout) | (None, 50)     | 0       |
| dense (Dense)       | (None, 20)     | 1020    |
| dropout_3 (Dropout) | (None, 20)     | 0       |
| dense_1 (Dense)     | (None, 10)     | 210     |
| dropout_4 (Dropout) | (None, 10)     | 0       |
| dense_2 (Dense)     | (None, 1)      | 11      |

=====  
Total params: 72,641  
Trainable params: 72,641  
Non-trainable params: 0



# Summary of Lesson 2

In Lesson 2, we have:

- ▶ Developed our first LSTM Neural Network model
- ▶ Performed Explanatory Data Analysis (EDA) on high-frequency data such as BTC prices
- ▶ Evaluated the predictive power of LSTM and backtest the strategy in test sample.

**TO DO NEXT:** Go to the accompanying Python Notebook to implement your first LSTM!

In the next lesson, we will learn more about LSTM by introducing bidirectional networks.